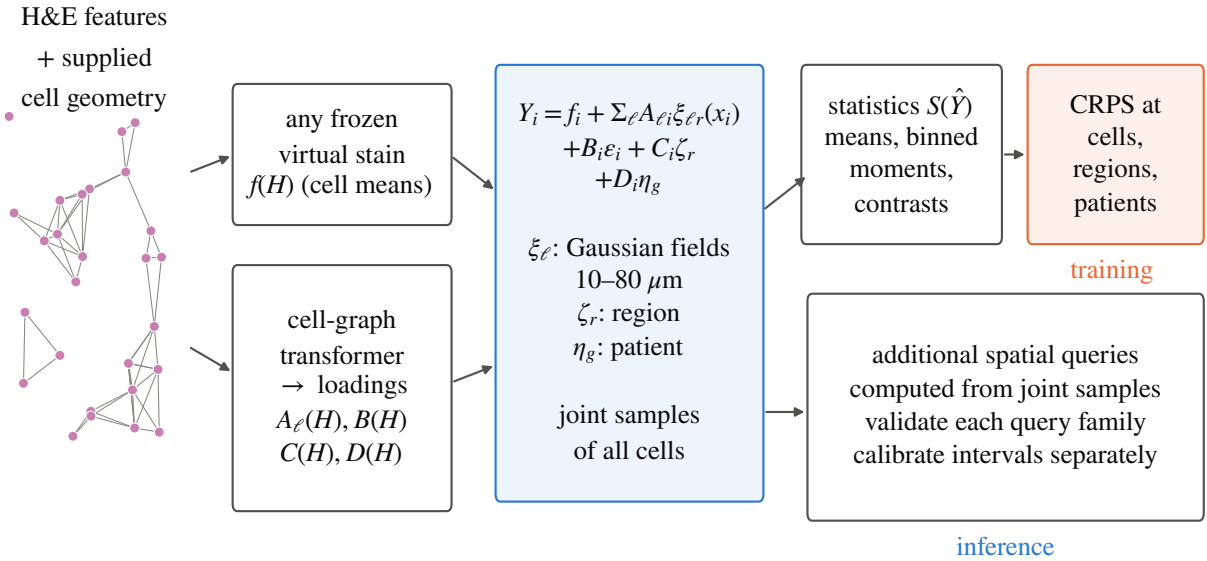
H&E features
+ supplied
cell geometry
any frozen
virtual stain
$f(H)$ (cell means)
cell-graph
transformer
→ loadings
$A_\ell(H), B(H)$
$C(H), D(H)$
$Y_i = f_i + \Sigma_\ell A_{\ell i}\xi_{\ell r}(x_i)$
$+B_i\varepsilon_i + C_i\zeta_r$
$+D_i\eta_g$
ξ_ℓ: Gaussian fields
10–80 μm
ζ_r: region
η_g: patient
joint samples
of all cells
statistics $S(\hat{Y})$
means, binned
moments,
contrasts
CRPS at
cells,
regions,
patients
training
additional spatial queries
computed from joint samples
validate each query family
calibrate intervals separately
inference